

Product Carbon Footprint Study Report

Product Name: kerosene production

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: kerosene production

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg kerosene

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg kerosene.

2. System Boundary

kerosene production, petroleum refinery operation, in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of kerosene production produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **0.801 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

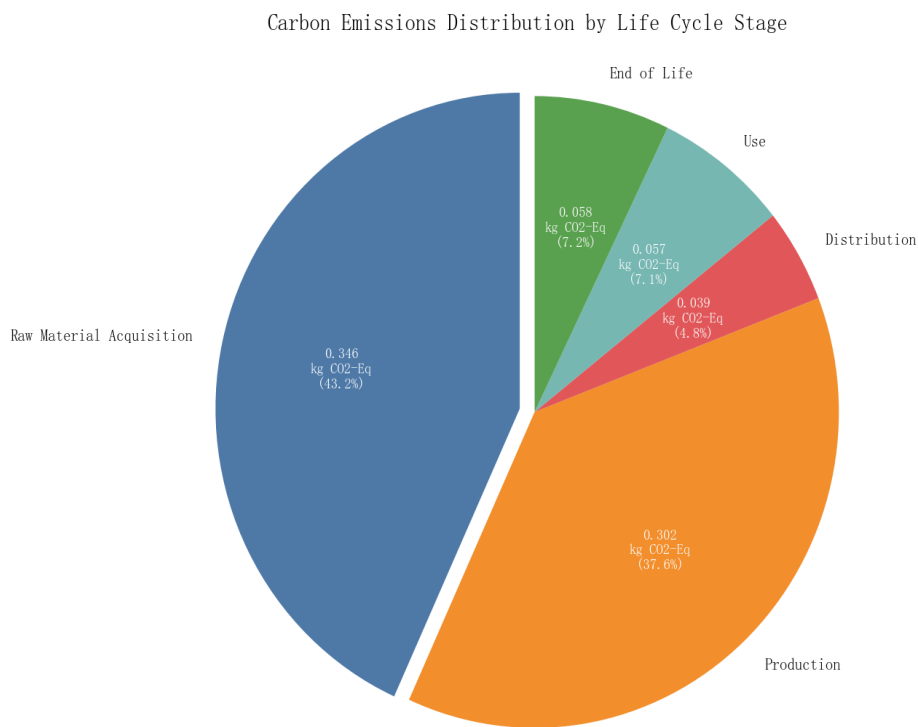


Table 1: Carbon Emissions by Life Cycle Stage

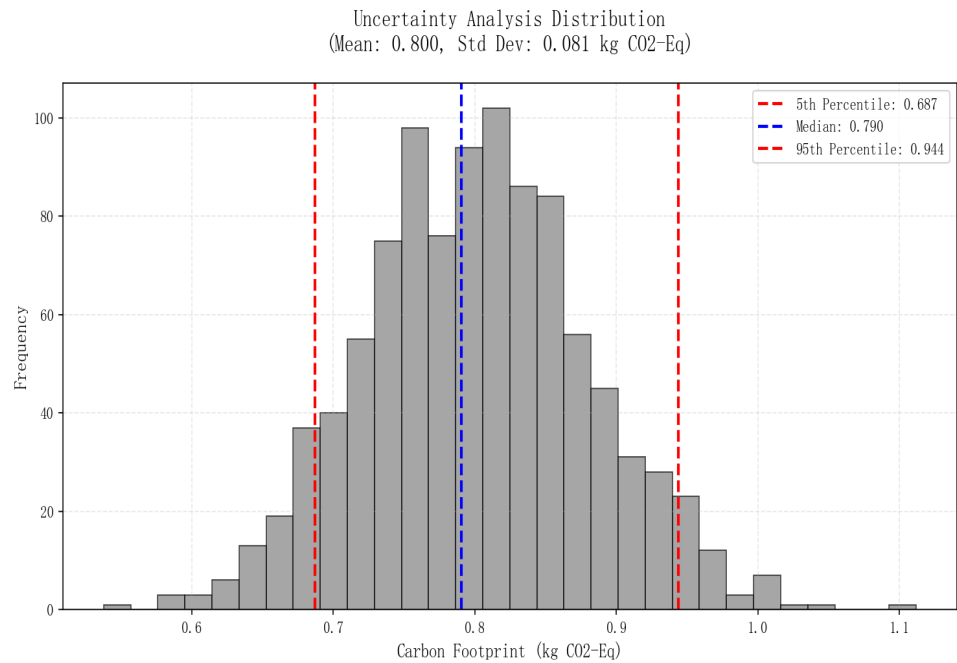
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	0.346	43.23
Production	0.302	37.64
Distribution	0.039	4.81

Use	0.057	7.12
End of Life	0.058	7.20
Total	0.801	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	0.800
Standard Deviation	0.081
Coefficient of Variation (CV)	10.12%
Median	0.790
5th Percentile	0.687
95th Percentile	0.944

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of kerosene production is quantified at 0.801 kg CO₂-Eq per functional unit, representing a moderate emission intensity typical for fossil fuel-derived products in industrial LCA studies. This value encapsulates the full lifecycle impacts and highlights kerosene's contribution to greenhouse gas emissions within energy supply chains.

Lifecycle stage contributions reveal distinct patterns, with raw_material extraction and production dominating at 43.2% and 37.6% respectively, collectively accounting for over 80% of total emissions. Distribution, use, and end_of_life phases contribute minimally, each below 8%, indicating that upstream processes drive the majority of impacts while downstream activities remain secondary.

Major emission sources are identified in raw_material acquisition, likely from crude oil extraction and refining inputs, and the production phase involving energy-intensive conversion processes. These hotspots underscore the need for targeted interventions in fossil feedstock sourcing and manufacturing efficiency to achieve meaningful decarbonization.

2. Statement of Assumptions and Limitations

Key assumptions in this study include the use of generic background datasets for upstream processes and an average functional unit based on standard kerosene output without site-specific variations. It is assumed that all stages follow a consistent allocation method for co-products and that emission factors reflect current technological averages rather than future projections.

Limitations of data sources and quality arise from reliance on secondary databases such as Ecoinvent or GaBi, which may introduce regional or temporal inaccuracies, particularly for raw_material extraction in diverse global locations. Primary data gaps could affect precision in production energy consumption, and variability in end-of-life scenarios remains underrepresented due to limited empirical measurements.

System boundary restrictions encompass a cradle-to-grave approach limited to the defined stages, excluding indirect effects such as land-use changes or infrastructure maintenance. The analysis does not extend to downstream applications beyond kerosene combustion or account for potential biogenic carbon credits, thereby constraining the scope to direct process-related emissions.

3. Suggestions for Improvement

- Target raw_material stage by sourcing lower-emission feedstocks or implementing carbon capture during extraction to reduce the 43.2% contribution.
- Optimize production efficiency through renewable energy integration and process heat recovery, directly addressing the 37.6% emissions from this dominant phase.
- Explore alternative distribution logistics, such as localized refining, to further minimize the smaller but cumulative transport-related impacts.

- Enhance data quality by collecting primary site-specific measurements for energy and material flows, supplemented with uncertainty analysis in future iterations.
- Pursue circular economy pathways, including end_of_life recycling of by-products, to lower overall lifecycle burdens across multiple stages.